

Prangan Nandy

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ACADEMIC CREDENTIALS

Ph.D. in Mechanical Engineering <i>University of South Carolina</i>	Jan. 2024 - Present <i>Columbia, SC, USA</i>
M.S. in Mechanical Engineering <i>University of South Carolina</i>	Aug. 2026 (Expected) <i>Columbia, SC, USA</i>
B.S. in Mechanical Engineering <i>Bangladesh University of Engineering and Technology (BUET)</i>	Feb. 2017 - May 2022 <i>Dhaka, Bangladesh</i>

PROFESSIONAL EXPERIENCE

Graduate Research Assistant - RASAER Laboratory <i>University of South Carolina</i>	Jan. 2024 - Present <i>Columbia, SC, USA</i>
- Conducted experimental investigations of atmospheric-pressure non-thermal plasmas, including piezoelectric transformer (PT) driven discharges and nanosecond-pulsed water vapor plasmas, through the development of diagnostic techniques, reactor configurations, and gas handling systems. - Performed advanced modeling and simulations to investigate piezoelectric transformer driven discharges, plasma assisted chemical conversion, plasma surface interactions, and plasma plume evolution in non-homogeneous gas mixtures - Supervised undergraduate students in laboratory research activities, including literature review, experimental preparation, data processing, and research-presentation development - Contributed to multi-institution plasma research collaborations with Sandia National Laboratories, the University of Notre Dame, and the University of Michigan by integrating simulations, diagnostics, and mechanistic interpretation across teams	
Assistant Engineer <i>Indetechs Software Limited</i>	Mar. 2023 - Dec. 2023 <i>Dhaka, Bangladesh</i>
- Built Python-based simulation models for engineering systems, including an adiabatic cooling model to estimate outlet temperature and water consumption for geothermal power plant operations - Prepared standardized Environmental Product Declaration (EPD) and Safety Data Sheet (SDS) templates for material-passport database supporting sustainable product data management	
Technical Management Trainee <i>Abul Khair Steel Products Limited</i>	Jul. 2022 - Feb. 2023 <i>Chittagong, Bangladesh</i>
- Conducted root cause analysis of mechanical failures of cold rolling mill, improving system reliability and operational stability - Designed a roll-changing mechanism using SolidWorks, reducing maintenance time and increasing operational profit by \$9000 per week	

RESEARCH EXPERIENCE

Unravelling the Coupled Physics of Piezoelectric and Plasma Behavior in Piezoelectric Stimulated Plasma Sources <i>Funded by: NSF ECLIPSE (Ecosystem for Leading Innovation in Plasma Science and Engineering)</i>	Jan. 2024 - Present
- Developed a fully coupled 2D multiphysics framework integrating piezoelectric mechanics and plasma kinetics to simulate PT-driven atmospheric-pressure helium discharges (a compact energy-efficient plasma source) - Designed experimental setup and executed synchronized voltage/current diagnostics and time-resolved ICCD imaging to characterize localized breakdown, polarity-dependent emission structures, and surface-guided plasma propagation - Interpreted experimental observations and predicted ionization pathways simultaneously to explain current pulses and discharge sustainment - Resolved underlying physics of PT surface discharges through predicted asymmetric charge-field interactions under different polarities	

- Developed a plasma-jet/surface-interaction model to predict how active piezoelectric substrates modify near-surface electric fields, surface charging, and charged-particle fluxes compared with passive glass substrates

Quantification of Reactive Species and Mechanistic Understanding of Plasma Assisted Siloxane Reformation Using Molecular Beam Mass Spectrometry (MBMS) Oct. 2024 – Dec. 2025

Funded by: Department of Energy

- Implemented a global plasma-chemistry model to explain HMDSO conversion products, fragmentation pathways, reactive intermediates, and power-dependent gas-phase clustering observed by molecular-beam mass spectrometry
- Supported mechanistic interpretation of the plasma-induced precursor formation along with early stages of PDMS generation at atmospheric pressure

Nonthermal H₂ plasma-enabled lignin hydrogenolysis to valuable chemicals and bio-oils June 2024

Collaborative Plasma-Assisted Biomass Conversion Research

- Supported a study of renewable and sustainable chemical conversion process by measuring discharge current and plasma power which enabled correlation between input power, reaction conditions, lignin conversion, and product-distribution trends

Advanced Adiabatic Compressed Air Energy Storage Apr. 2021 - May 2022

Undergraduate Thesis

- Developed a Python-based numerical model of an advanced adiabatic compressed air energy storage system integrated with rectangular thermal energy storage units, contributing to clean energy generation and storage modeling
- Evaluated effects of compression staging and air storage pressure on efficiency and exergy loss; results indicated higher storage efficiency with additional compression stages and reduced throttling exergy loss at higher storage pressure

PUBLICATIONS

1. **Prangan Nandy**, Md Fahim, Michael Tyrell, Jinyu Yang, David B. Go, Sourav Banerjee, and Tanvir Farouk. "Multiphysics Modeling and Diagnostics of Piezoelectric Transformers as a Plasma Source: Dynamic Polarity-Dependent Mechanism of Surface-guided Discharges." (**In review at Plasma Sources Science and Technology**)
2. Md Fahim, **Prangan Nandy**, David B. Go, Tanvir Farouk, and Sourav Banerjee. "Electric Field Localization in Piezoelectric Transformers Enables Atmospheric Pressure Plasma Generation at Low Voltage." (**In review at Smart Materials and Structures**)
3. Farouk, Tanvir I., **Prangan Nandy**, Raziya S. Chowdhury, Shamia Hoque, Emma C. Litzer, and Nils Hansen. "Siloxane Decomposition and Clustering in Atmospheric Pressure Non-Thermal Plasma Discharge." **Plasma Processes and Polymers**, 2026
4. Pishva, Parsa, Jialu Li, Rongxuan Xie, Jinyao Tang, **Prangan Nandy**, Tanvir Farouk, Jinghua Guo, and Zhenmeng Peng. "Nonthermal hydrogen plasma-enabled ambient, fast lignin hydrogenolysis to valuable chemicals and bio-oils." **Chemical Engineering Journal**, 2024

CONFERENCE PRESENTATIONS / POSTER

1. Hansen, Nils, Tanvir I. Farouk, **Prangan Nandy**, Shamia Hoque, Raziya Sultana Chowdhury, Oisin J. Shiels, and Emma C. Litzer. "Near-threshold photoionization molecular beam mass spectrometry for identifying reactive intermediates and products in an atmospheric pressure non-thermal plasma operating in siloxane." **78th Annual Gaseous Electronics Conference (GEC)**, Oct. 13-17, 2025, Seoul, Korea.
2. **Nandy, Prangan**, Md Fahim, Sourav Banerjee, and Tanvir Farouk. "Characteristics of PT-Driven Plasma Discharge at Atmospheric Pressure." **26th International Symposium on Plasma Chemistry (ISPC 26)**, June 15-20, 2025, University of Minnesota, MN.
3. Farouk, Tanvir, **Prangan Nandy**, Raziya Sultana Chowdhury, Shamia Hoque, Emma Litzer, and Nils Hansen. "Measurements of reactive species in an atmospheric pressure non-thermal plasma operating in siloxane: Mechanistic understanding." **77th Annual Gaseous Electronics Conference (GEC)**, Sept 30 – Oct 4, 2024, San Diego, CA.
4. **Nandy, Prangan**, Md Mushfiqur Rahman Fahim, Sourav Banerjee, and Tanvir Farouk. "Dynamics of a PT Driven Atmospheric Pressure Plasma Discharge." **77th Annual Gaseous Electronics Conference (GEC)**, Sept 30 – Oct 4, 2024, San Diego, CA.

SERVICE TO PROFESSION

- Reviewer, Physics of Plasmas (May 1, 2026)

HONORS & AWARDS

- Dean's Award, Bangladesh University of Engineering and Technology, 2021 and 2022.
- Government Education Board Scholarships, Bangladesh, awarded at primary, secondary, and higher-secondary levels.

TECHNICAL SKILLS

- **Simulation & Modeling:** COMSOL Multiphysics, ANSYS/ANSYS Fluent, STAR-CCM+, FactSage, plasma chemical-kinetics, numerical methods
- **Experimental & Diagnostics:** ICCD imaging, optical emission spectroscopy, voltage-current diagnostics, system fabrication
- **Programming & Data Analysis:** Python, MATLAB, OriginPro, Tecplot 360
- **CAD & Engineering Design:** SolidWorks (Certified SolidWorks Associate - Mechanical Design), AutoCAD.